

6.Create tensors using PyTorch and perform basic mathematical operations.

Code:

```
import torch

a = torch.tensor([1, 2, 3])
b = torch.tensor([4, 5, 6])

print("Tensor A:", a)

print("Tensor B:", b)

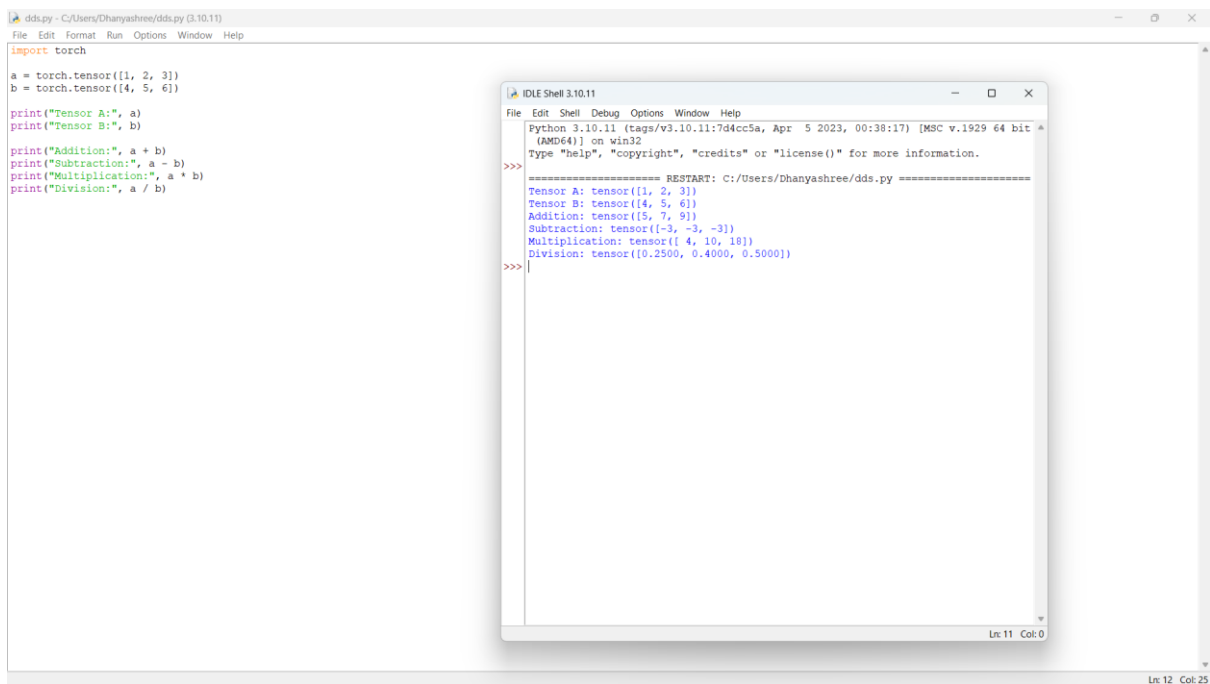
print("Addition:", a + b)

print("Subtraction:", a - b)

print("Multiplication:", a * b)

print("Division:", a / b)
```

OUTPUT

The image shows a screenshot of a Python IDE with two windows. The main window on the left displays the code from the previous block. The 'IDLE Shell 3.10.11' window on the right shows the execution output. The output starts with a restart message, followed by the tensor definitions for A and B, and then the results of the addition, subtraction, multiplication, and division operations. The division result is shown as a tensor with floating-point values.

```
Python 3.10.11 [tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17] [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/dds.py =====
Tensor A: tensor([1, 2, 3])
Tensor B: tensor([4, 5, 6])
Addition: tensor([5, 7, 9])
Subtraction: tensor([-3, -3, -3])
Multiplication: tensor([ 4, 10, 18])
Division: tensor([0.2500, 0.4000, 0.5000])
>>>
```

